

6(a)	1 (open top tap and) keep bottom tap closed / close bottom tap for a longer time (and then open bottom tap to collect product) A close bottom tap for complete hydrolysis to occur or run starch solution through column, collect product, repeat (a number of times) / AW or only allow a small quantity of starch solution through at a time / AW ; <i>if a different idea suggested to set-up in Fig. 6.1, method must be related to an increase in contact time between beads and starch to award mp1</i> any two from: 2 suggestion of modification to experimental set-up in Fig. 6.1 ; e.g. use smaller beads use more, beads / enzyme in beads I use more immobilised enzyme use of visking tubing attached to end of column (so only reducing sugar passes out) use longer, narrower column I ref. to filtering with filter paper / adding buffers to control pH 3 test (sample of) product with iodine solution, qualified ; A iodine in potassium iodide solution / I in KI solution e.g. to confirm no starch present stays orange (colour) does not change to blue-black (colour) to confirm no starch present 4 ref. to using optimum temperature (for amylase) ; A e.g. 35 °C temperature	3
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Question	Answer	Marks
6(b)	<i>any one from:</i> surface area to volume ratio / SA:V, decreases with increasing size / increases with decreasing size ; SA:V ratio is different, with consequence ; e.g. smaller beads means increased contact of substrate with enzyme (in same conditions) <i>idea that overall, larger surface area exposed using smaller beads ; ora</i> <i>(for the same experimental set up)</i> larger beads, slows rate of diffusion into beads / increases diffusion time to reach enzyme ; ora larger beads will increase rate of flow through column ; ora	1